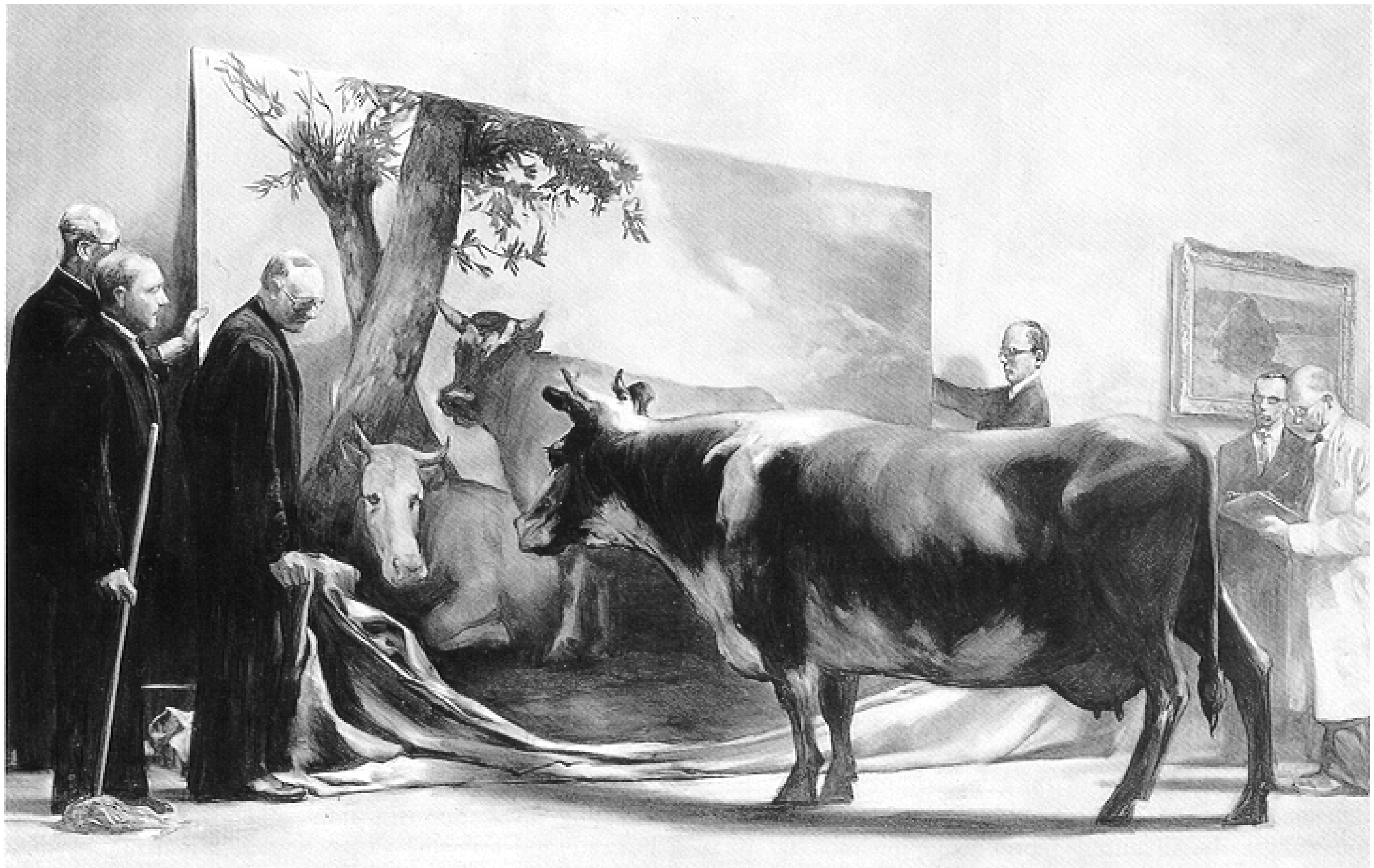
ENVX2001: Applied Statistical Analysis



Lecture 4 – Cluster analysis

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Outline

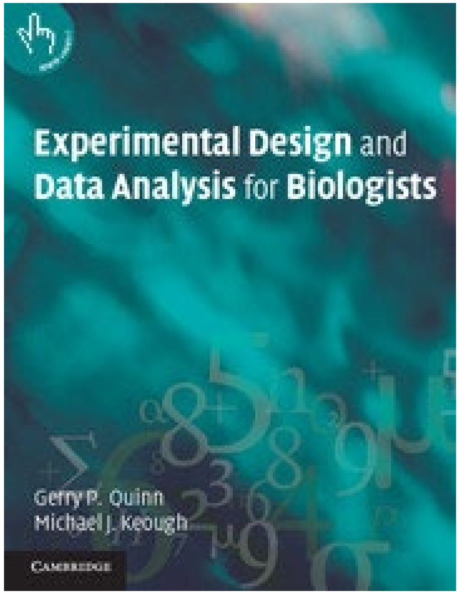
- Multivariate data and
- Hypothesis testing and data exploration*
- Important techniques including
  - PCA
  - Cluster analysis*
  - nMDS
  - Monte Carlo approaches
  - MANOVA

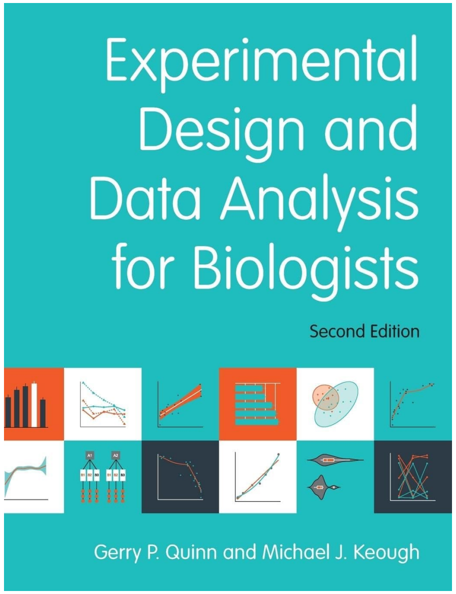
Lectures 3, 4, 5

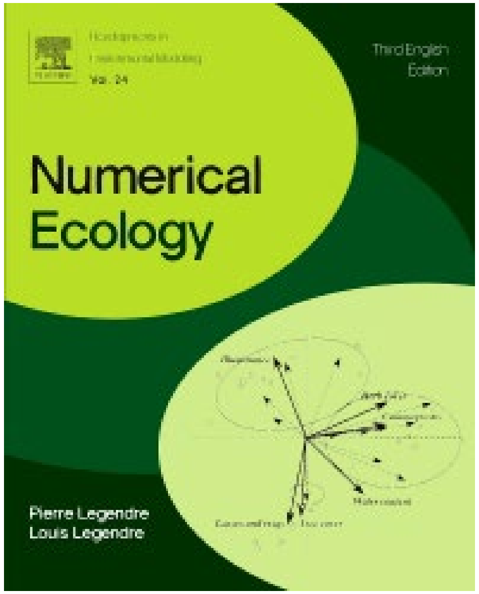
- Distance matrices
- Transformations and standardisations
- *Cluster analysis
  - Types of questions
  - Carrying it out
  - Interpreting it
- Hypothesis testing (and best practice)
- Steps for testing
  - Visualising (nMDS)
  - Randomization (generating *p* using ANOSIM and PERMANOVA)
  - Post-hoc tests
  - Identifying contributors to the differences (SIMPER)

References

- Quinn and Keough (2002) – Chapter 18
- Quinn and Keough (2023) – Chapter 16
- Legendre & Legendre (2012) - Chapter 8
- .Han *et al.* (2023)





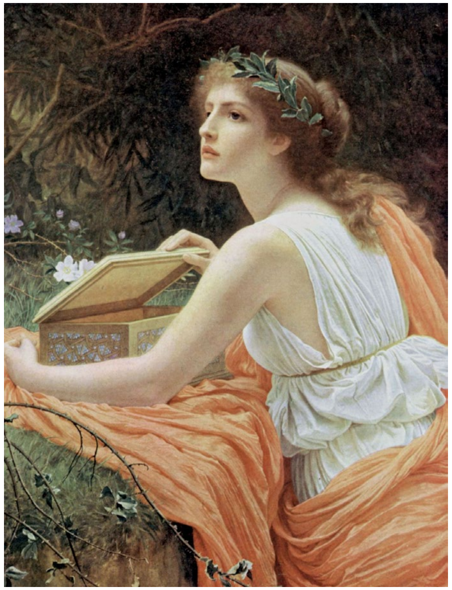


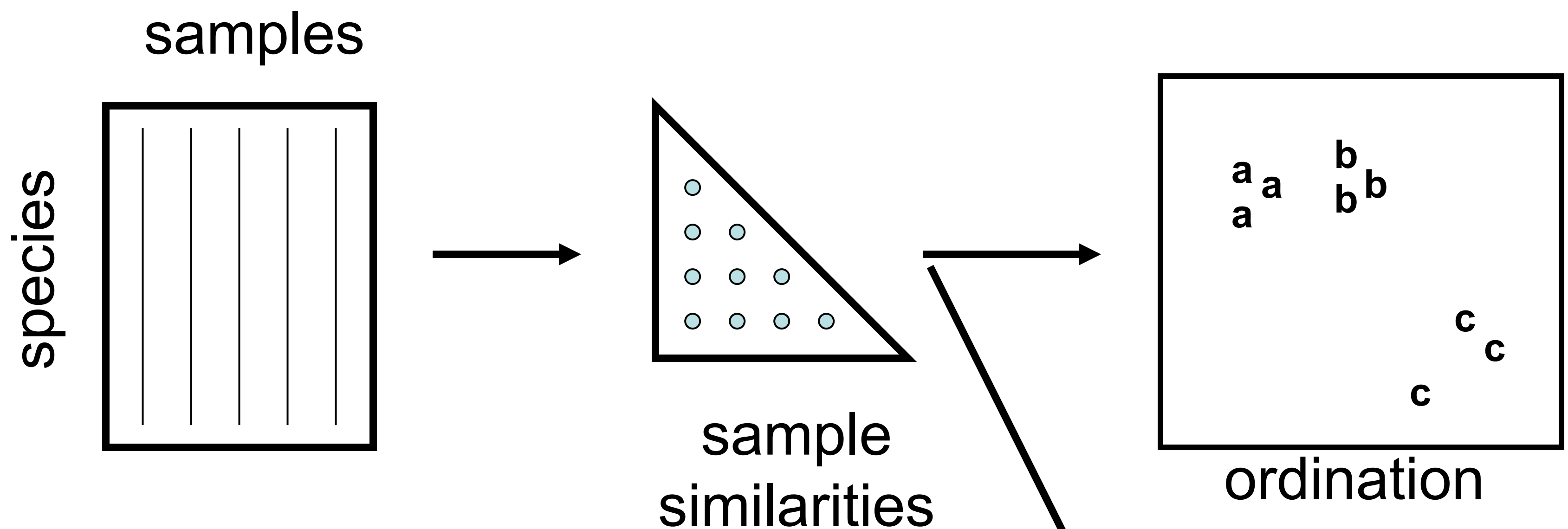
Using similarity matrices

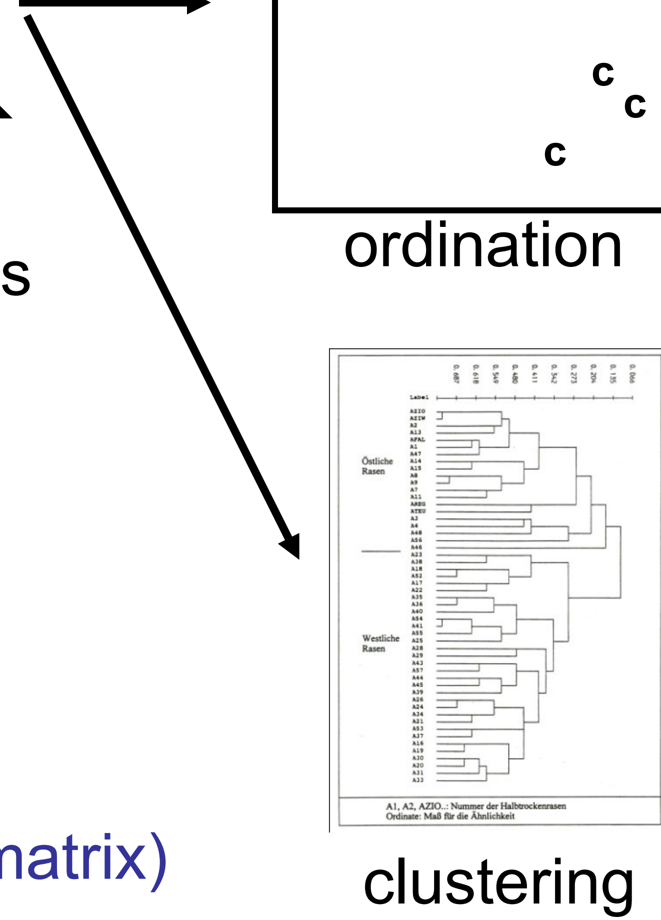
- Cluster analysis
  - Types of questions
  - Carrying it out
  - Interpreting it
- Hypothesis testing (and best practice)
- Steps for testing
  - Visualising (nMDS)
  - Tests (generating *p* using ANOSIM/PERMANOVA)
  - Post-hoc tests
  - Identifying contributors to the differences (SIMPER)

Multivariate approaches

- Assumption laden and highly accessible - “Pandora’s box or panacea”
- Based on extent to which samples share a species at comparable abundance
- All use similarity coefficients calculated between each pair of samples
- Major Approaches
  1. Clustering into groups
  2. Ordination (Mapped)







The story so far…

Assuming:

- Multivariate data
- Transformations*
- Standardisations*
- Similarity matrix (or dissimilarity matrix)

Clustering - the question

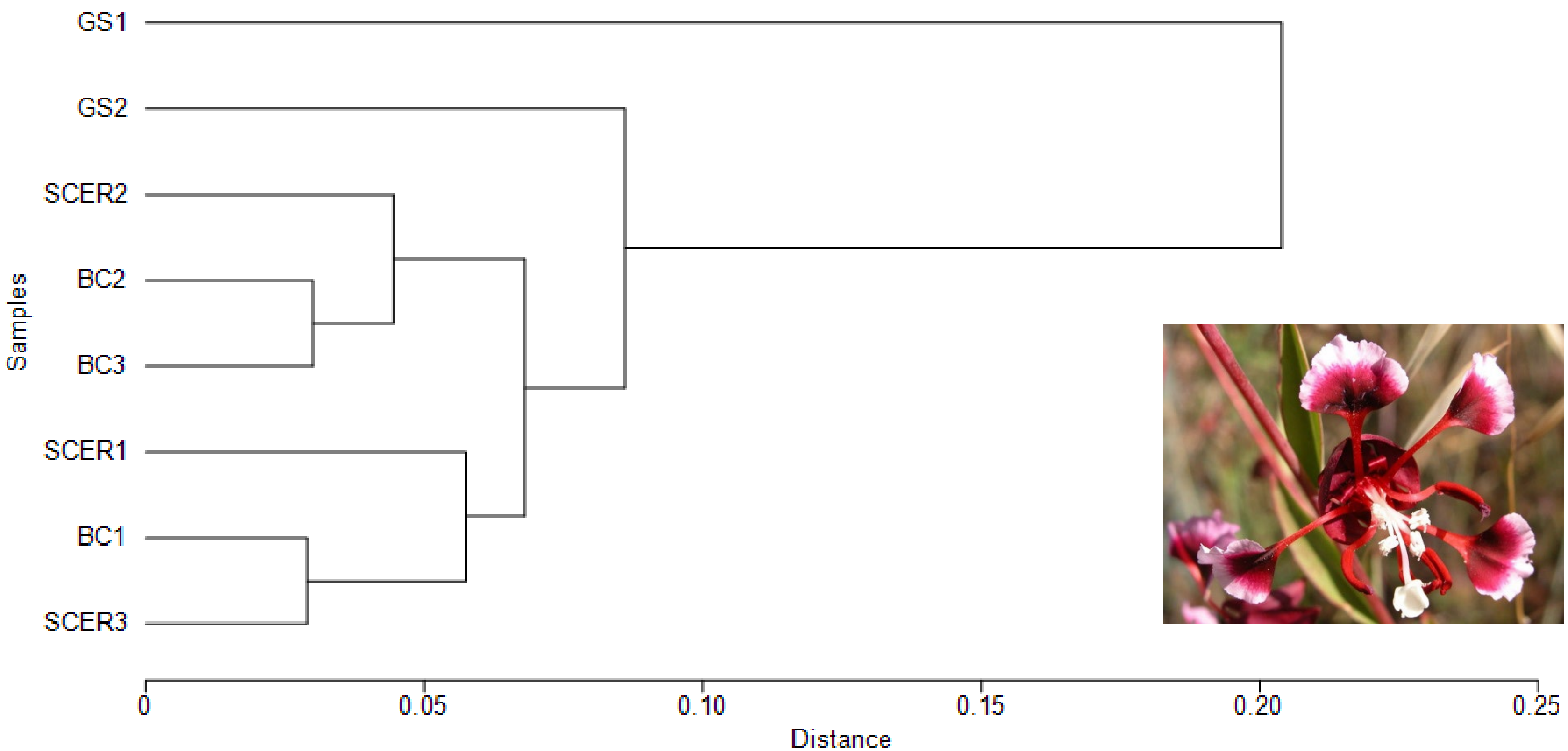
- Do samples associate with each other into groups?
- Consider
  - Finding groupings among “species” generated by using morphometric or genetic data
  - Looking for groupings among sites based species, soil characteristics etc

Genetic Structure within Rare Plant

Rare annual plant from California (*Clarkia springvillensis*)

- Three main populations with subpopulations Gauging Station (GS), Springville Clarkia Ecological Reserve (SCER) and Bear Creek (BR)
- Used Cavalli-Sforza chord genetic distances

Genetic Structure within Rare Plant

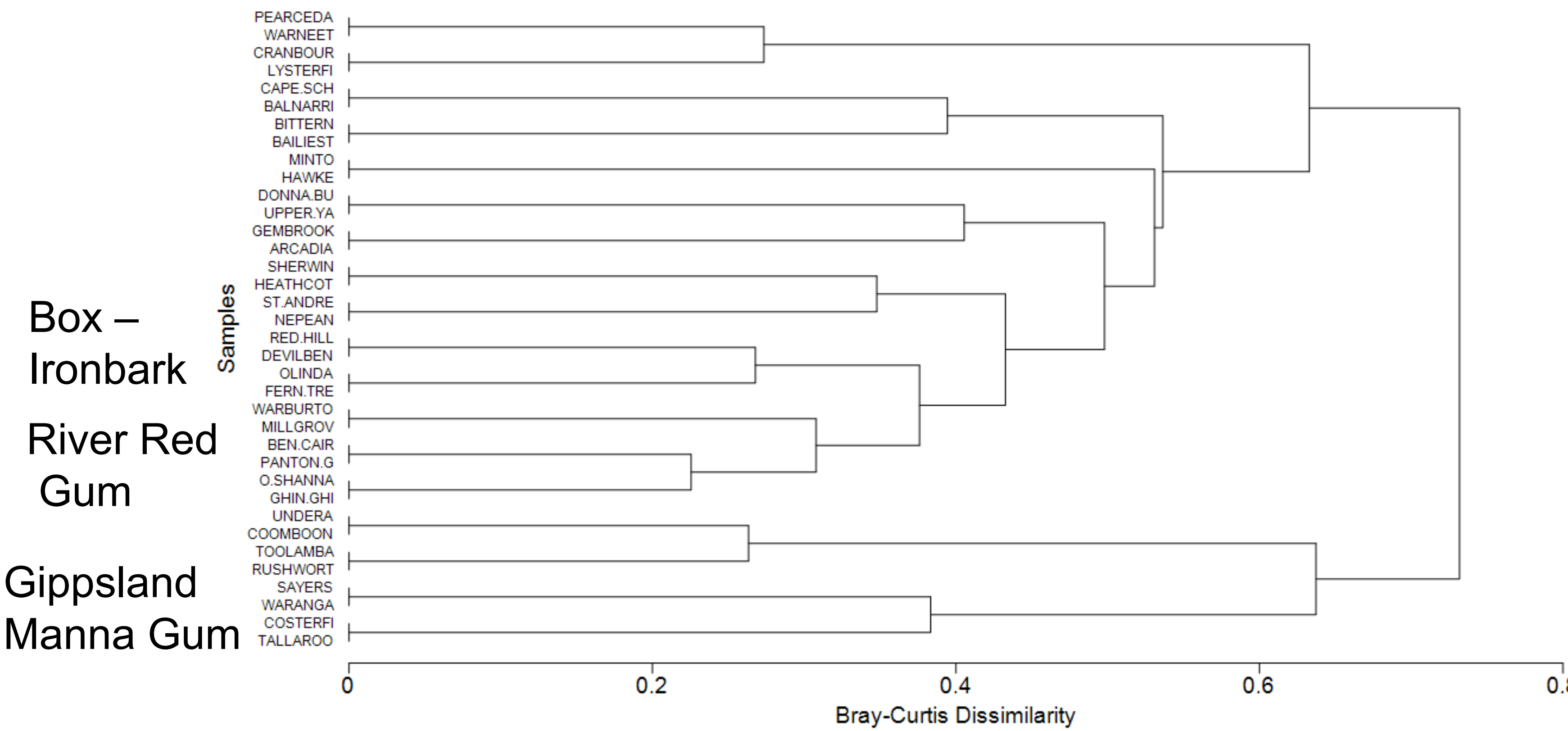


McCue *et al.* (1996)

Patterns of bird assemblages in southeastern Australia

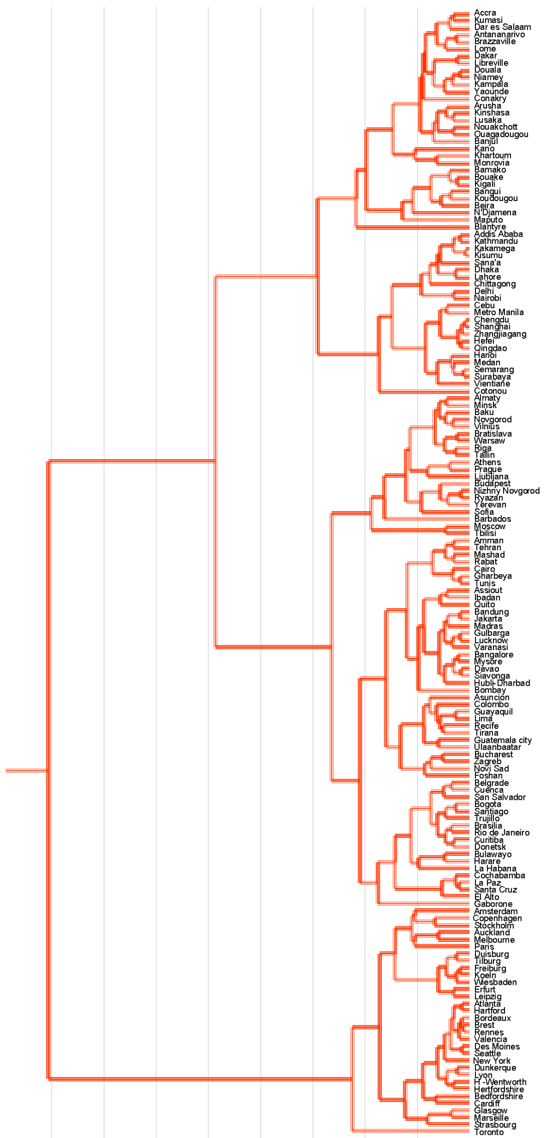
- Birds counted in different vegetation types in Victoria
- Is there a pattern in bird assemblages for different vegetation type?

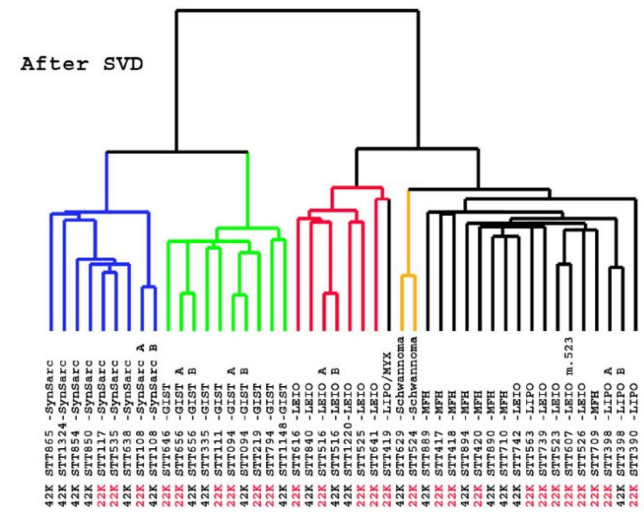
Patterns of bird assemblages in southeastern Australia



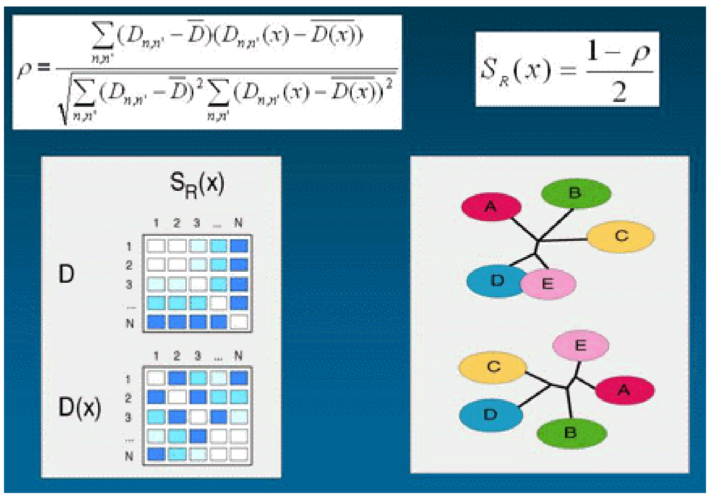
Mac Nally (1989)

Clustering - the output





Webfigure 5a

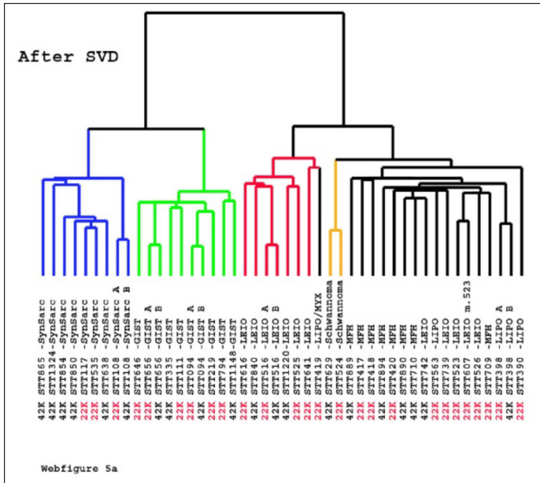


Clustering (classifying)

- aims to find "natural groupings"
- samples within a group are more similar to each other than samples in different groups
- Used a variety of ways, usually resulting in a dendrogram
- Imposing a structure on the data or revealing the structure which actually exists?
- Exploratory data analysis/hypothesis generation
- Common in taxonomy, less so in ecology
- Method must be numerical and **the final number of classes is not known**
  - c.f. discriminant function analysis

Clustering - the process/meaning

- Generate distance matrix
- Choose clustering approach
  - – Agglomerative or divisive
  - – Hierarchical or non hierarchical
  - – Weighting mean differences or not weighting
- Look for groupings



Ways of doing cluster analysis

- hundreds of methods – do not need to know all
- Most methods are sequential
- Application of a sequence of operations that is either disjoint (agglomerative) or conjoint (divisive)
- Hierarchical and non-hierarchical
- Cluster analysis usually "agglomerative hierarchical" in ecology and systematics – agriculture uses more non-hierarchical

Hierarchical Agglomerative Cluster Analysis

- start with a pairwise similarity matrix among objects (individuals, sites, populations, taxa)
- Most similar joined into a group
- Similarities of new groups to all others is calculated
- Two closest groups are combined repeatedly until one group remains
- 2 dimensional representation in dendrogram

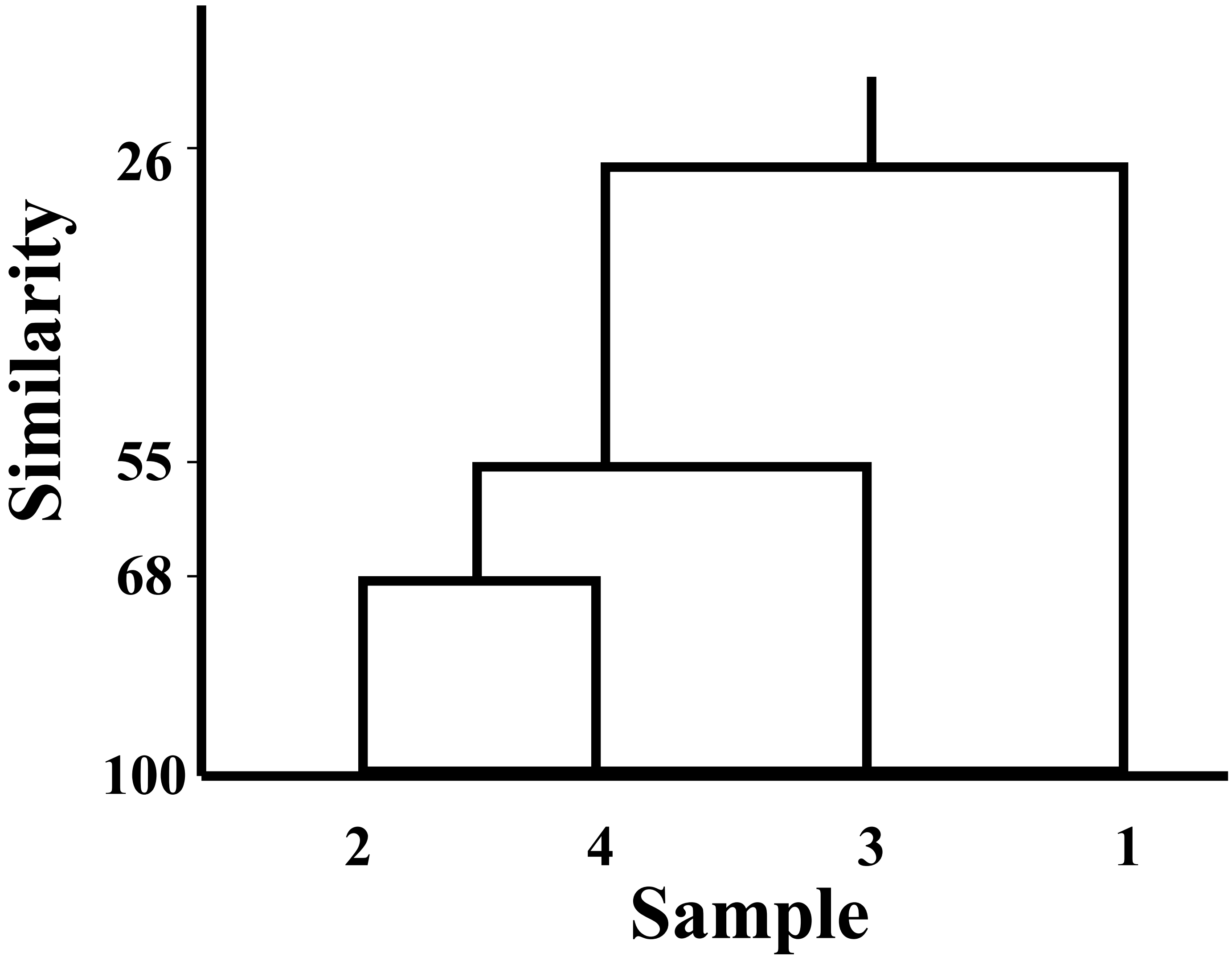
Types of Hierarchical Agglomerative Cluster Analysis

- Single Linkage (nearest neighbour)
- Complete Linkage (furthest neighbour)
- Average Linkage (group average or mean)
  - Unweighted pair-groups method using arithmetic averages (UPGMA) – common!
  - Weighted pair-groups method using arithmetic averages (WPGMA)
  - Unweighted pair-groups method using arithmetic centroids (UPGMC)
- Ward’s Minimum Variance Clustering

Transformed Abundance array (from yesterday's lecture)

Sample	1	2	3	4
Sp. 1	1.7	0	0	0
Sp. 2	2.1	0	0	1.3
Sp. 3	1.7	2.5	0	1.8
Sp. 4	0	1.9	3.5	1.7
Sp. 5	0	3.4	4.3	1.2
Sp. 6	0	0	0	0

Interpreting a dendrogram?
What's a meaningful grouping?



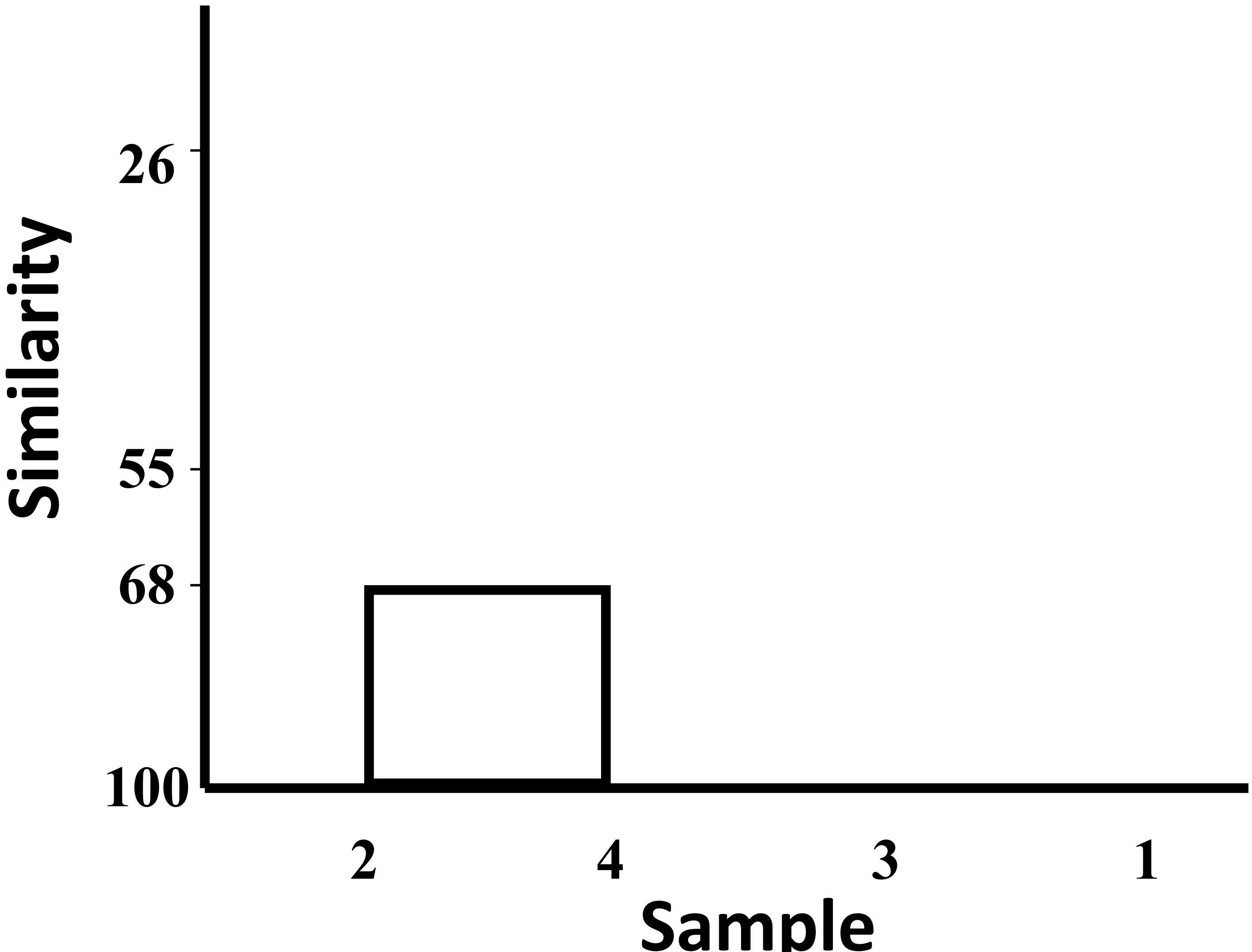
From similarity (Bray-Curtis) to clusters (UPGMA)

Sample	1	2	3	4
1	-			
2	25.6	-		
3	0.0	67.9*	-	
4	52.2	$\boxed{68.1}$	42	-

Note: Don't need to know calculations for exam or prac – R does it for you

((2&3) + (3&4))/2= (67.9+42)/2 = 55

Interpreting a dendrogram?
What's a meaningful grouping?



How similar are the last two clusters?

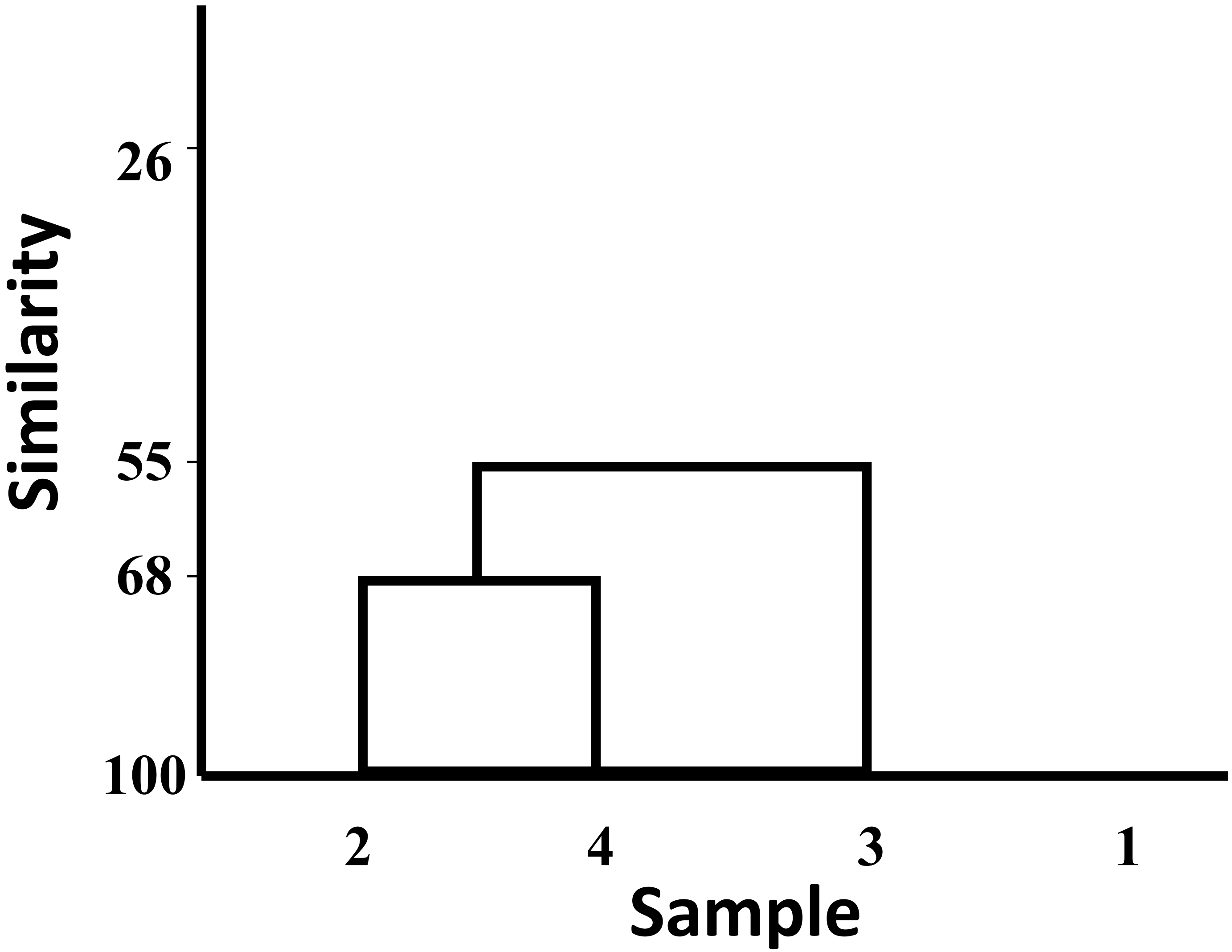
Sample	1	2&4	3
1	-		
2&4	38.9	-	
3	0.0	55.0	-

(2*(1,2&4) + (1&3))/3 = (2*38.9 +0)/3

(2*38.9+0)/3 = 25.9

"proportional averaging"

Interpreting a dendrogram?
What's a meaningful grouping?

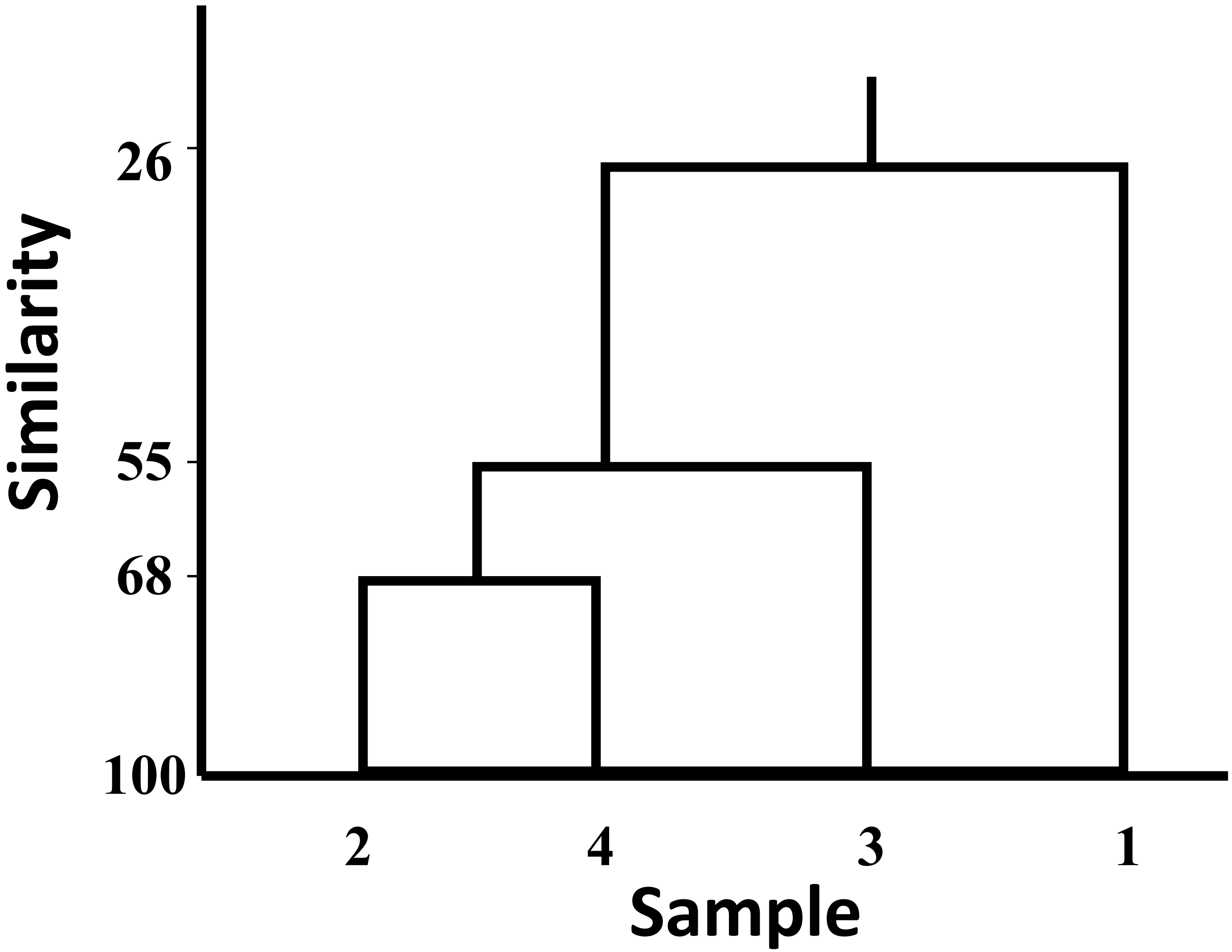


Similarity to last group?

Sample	1	2&3&4
1	-	
2&3&4	25.9	-

UPGMA – “unweighted paired group method using arithmetic averages”

Interpreting a dendrogram?
What's a meaningful grouping?



How to do hierarchical clustering in R

```
# find distance matrix (i.e. Bray-Curtis)
d <- dist(as.matrix(data))
# apply hierarchical clustering
 hc <- hclust(d)
# plot the dendrogram
plot(hc)
```

Non –hierarchical clustering

- Start with single object and cluster other objects that are similar to that one
- Objects can be reassigned to clusters during clustering process

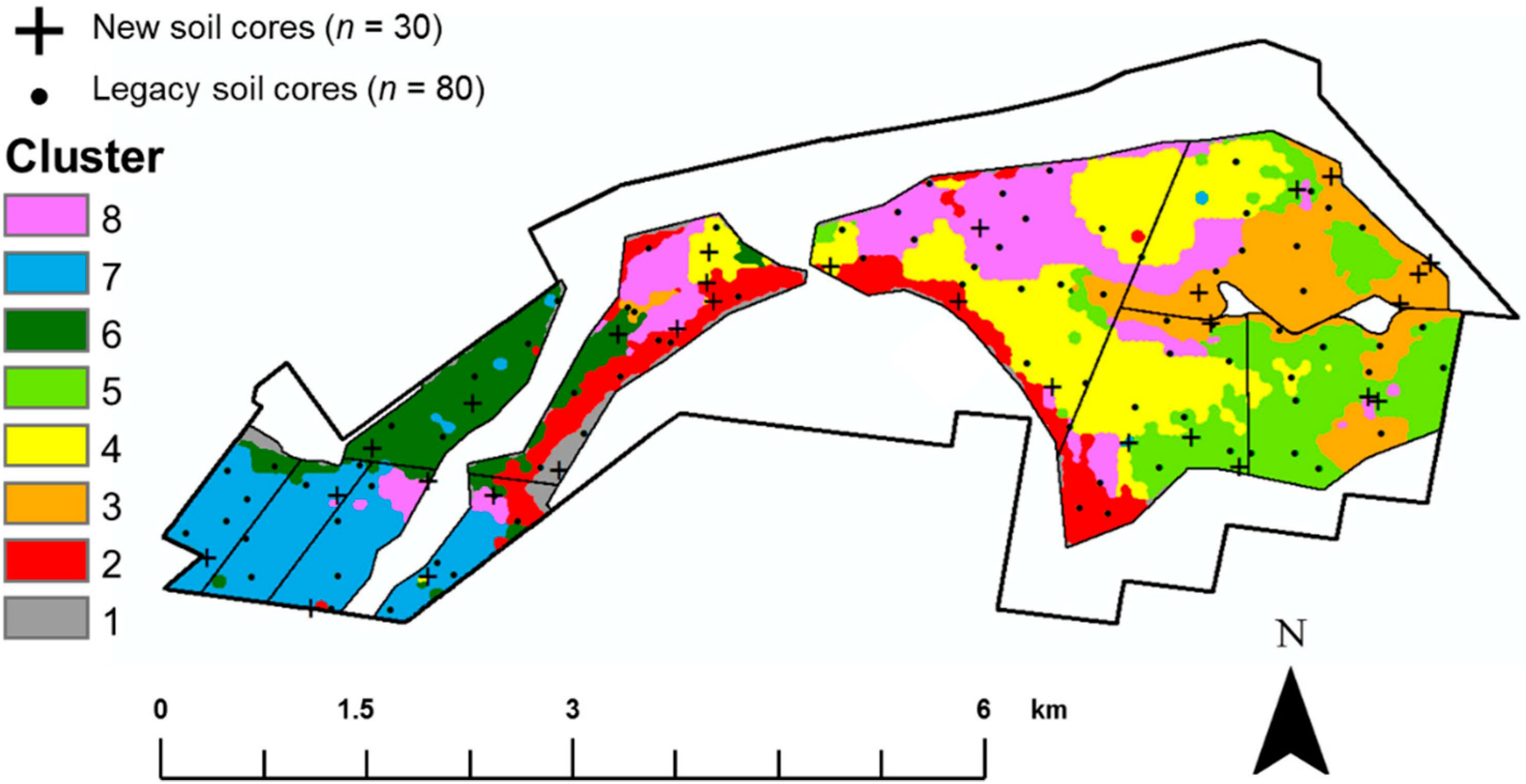
K-means clustering

- A type of non-hierarchical clustering
- Splits objects into pre-defined number (*K*) of clusters
- Cluster membership is iteratively re-evaluated by some criterion

How to do K-means clustering in R?

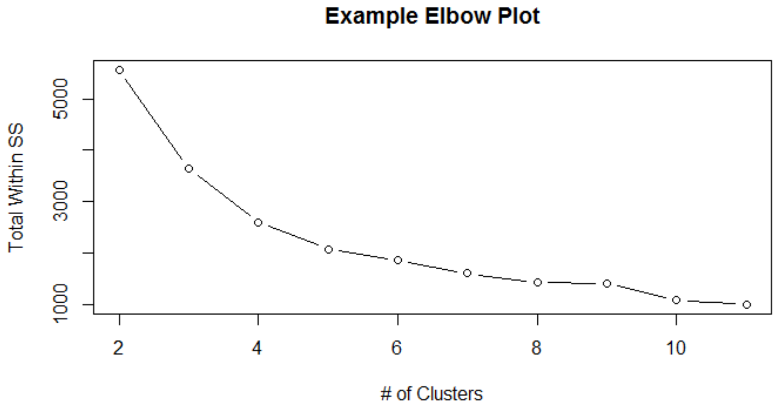
```
irisCluster <- kmeans(iris[, 3:4], 3, nstart = 20)

table(irisCluster$cluster, iris$Species)

irisCluster$cluster <- as.factor(irisCluster$cluster)

ggplot(iris, aes(Petal.Length, Petal.Width, color = irisCluster$cluster)) + geom_point()
```

L’lara farm soil classified by 8 clusters



How many clusters?

- By looking at a plot of the number of clusters x the Weighted Sum of Squares (dispersion in the groups around their respective means) – like a Scree Plot
- Calinski-Harabasz criterion
- Many others!



How many clusters in R?

```
mydata <- iris[,3:4]
wss <- (nrow(mydata)-1)*sum(apply(mydata,2,var)) for (i in 2:15) wss[i] <- sum(kmeans(mydata, centers=i,nstart=20)$withinss)
plot(1:15, wss, type="b", xlab="Number of Clusters", ylab="Within groups sum of squares“
# Intepret like screeplot - see lab
```

Cluster analysis – a summary

- Cluster analysis may be useful in a number of circumstances (e.g. taxonomy) BUT there are generally better options for environmental scientists
- Clustering is less useful (and potentially misleading) when there is steady gradation across sites/variables of interest
- Even for strongly grouped samples there are other ways of representing groups
- Ordination and hypothesis tests better options for well designed work